

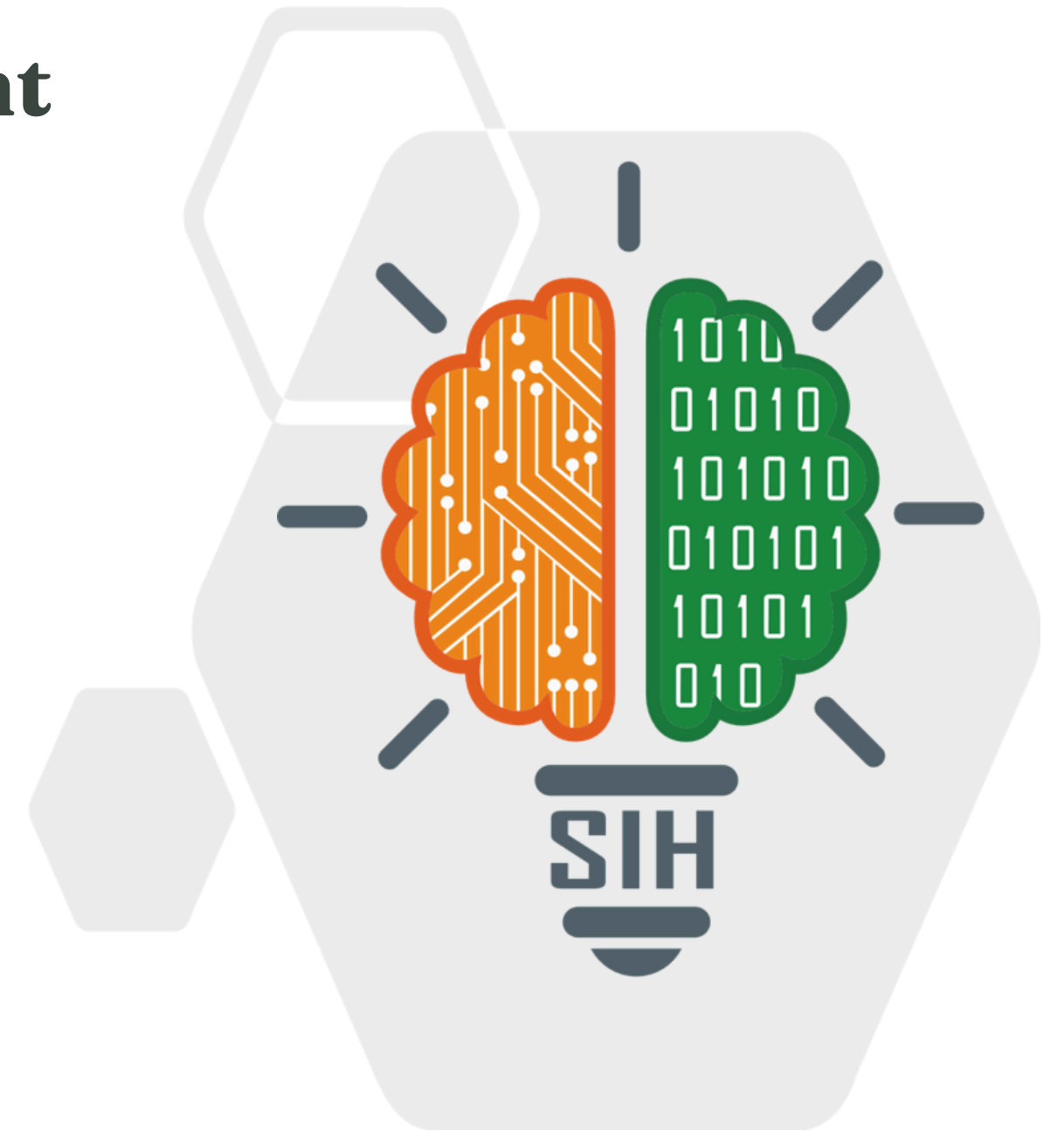
Multilingual RAG-Based Ayurveda IPR Assistant

Team: VedaVanguard

Problem Statement ID: SIH26045

Theme: Healthcare / Traditional Knowledge

Category: Software



Proposed Solution

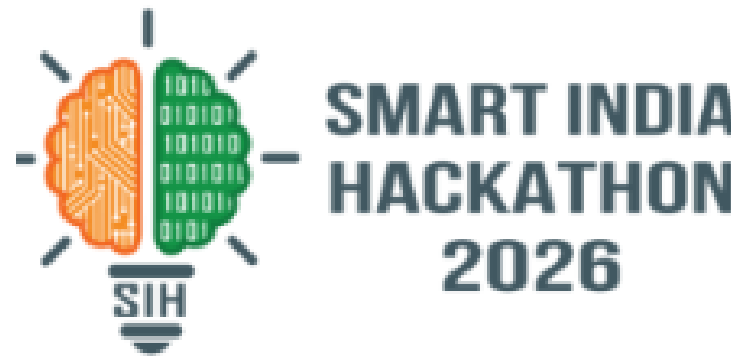
Multilingual RAG assistant for Ayurveda IPR (English, Hindi, Sanskrit, regional languages)

Jurisdiction-aware guidance (India + international: TKDL, Patent Act, WIPO, Nagoya Protocol)

Formulation classifier with ABS checks (Classical, Proprietary, Single/Polyherbal)

Source-grounded answers only with safe abstention on uncertain queries First RAG system of its kind combining multilingual Ayurveda knowledge with IPR logic

Technical Approach



User Input

Multilingual query + jurisdiction
toggle (India / International)

RAG Retrieval

Hybrid dense + BM25 over
curated legal corpus in
ChromaDB with
jurisdiction
metadata filtering

Guardrailed Generation

Citation-grounded answer,
confidence score, safe
abstention, disclaimer,
DPDP audit log

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Formulation Classifier

Pre-RAG router: Classical /
Proprietary / New Drug /
Phytopharma / Cosmetic

Compliance Modules

ABS helper (PIC/MAT
checklists), TKDL prior-art
mapper, Knowledge Graph
multi-hop

Feasibility & Viability

- **Zero research risk —mature stack: FastAPI, ChromaDB, MiniLM-L12-v2, Gemini/Groq, Bhashini APIs**
- **Corpus is open & licensable — Patent Act, Biodiversity Act, GI Registry, API, TKDL, WIPO GRATK 2024, Nagoya — all government/public sources**
- **Multilingual solved via Bhashini — MeitY's 22-language translation/ASR/TTS APIs ready; free tier, REST integration**
- **Shippable in 6–8 weeks by 2–3 devs — no GPU, serverless-friendly (~\$30/mo), stateless horizontal scale**
- **Clear business + policy tailwind — 50K+ AYUSH/MSME users, SaaS/API/white-label revenue, MeitY/BIRAC/DST grants, 2024 rules make compliance mandatory**

Impact & Benefits

Social Impact:

- **Protects traditional knowledge from biopiracy & wrongful patents**
- **Empowers Ayurveda practitioners, researchers, startups with instant IPR guidance**
- **Bridges language barrier — Sanskrit/Hindi knowledge accessible globally**
- **Supports AYUSH Ministry's digital India & traditional medicine globalization goals**

Economic Impact:

- **Reduces IP litigation costs for Indian pharma/nutraceutical companies**
- **Enables compliant export via ABS/Nagoya Protocol readiness checks**
- **Opens licensing revenue for classical formulation holders via GI tagging**

Environmental Impact:

- **Promotes sustainable bio-resource use via ABS compliance**
- **Discourages exploitative bioprospecting through prior-art transparency**

Research & References



- **Traditional Knowledge Digital Library (TKDL)—CSIR&Ministry of AYUSH**
- **India Patent Act, 1970 — Section 3(p) (traditional knowledge exclusion)**
- **Biological Diversity Act, 2002 & Access and Benefit Sharing Regulations, 2014**
- **WIPO Intergovernmental Committee on Intellectual Property & Genetic Resources, Traditional Knowledge & Folklore**
- **Nagoya Protocol on Access to Genetic Resources & Fair/Equitable Benefit Sharing**
- **TRIPS Agreement — Article 27.3(b) (patent exclusions for plants/animals/traditional knowledge)**
- **Charaka Samhita, Sushruta Samhita, Ashtanga Hridaya — Classical Ayurveda texts**
- **CCRAS (Central Council for Research in Ayurvedic Sciences) Pharmacopoeia & Formulary**
- **Geographical Indications Registry — India (e.g., Kandhamal Haladi, Assam Orthodox Tea)**
- **Sentence-Transformers: Reimers & Gurevych (2019) — Multilingual MiniLM models**
- **ChromaDB: Open-source vector database for LLM applications**
- **Groq Cloud: High-throughput LLM inference (Qwen, Llama, Mixtral models)**